

PRACTICE SHEET

1. In a triangle ABC, $a = 2b$ and $\angle A = 3\angle B$. Which one of the following is correct?
 - (a) The triangle is isosceles
 - (b) The triangle is equilateral
 - (c) The triangle is right – angled
 - (d) Such triangle does not exist
2. The median AD of a triangle ABC is bisected at F, and BF is produced to meet the side AC in P. If $AP = \lambda AC$, then what is the value of λ ?
 - (a) $\frac{1}{4}$
 - (b) $\frac{1}{2}$
 - (c) $\frac{2}{3}$
 - (d) $\frac{1}{3}$
3. If the perimeter of a triangle ABC is 30cm, then what is the value of $a \cos^2(C/2) + c \cos^2(A/2)$?
 - (a) 15cm
 - (b) 10cm
 - (c) $\frac{15}{2}$ cm
 - (d) 13cm
4. In $\triangle ABC$, if $\angle A : \angle B : \angle C = 1 : 2 : 3$, then what is $BC : CA : AB$?
 - (a) $1 : 2 : 3$
 - (b) $1 : \sqrt{3} : 2$
 - (c) $2 : \sqrt{3} : 1$
 - (d) $\sqrt{3} : 1 : 2$
5. The angles A, B, C of a triangle are in the ratio $2 : 5 : 5$. What is the value of $\tan B \tan C$?
 - (a) $4 + \sqrt{3}$
 - (b) $4 + 2\sqrt{3}$
 - (c) $7 + 4\sqrt{3}$
 - (d) $3 + 3\sqrt{3}$
6. In a triangle ABC, if $a = 2b$ and $A = 3B$ then which one of the following is correct?
 - (a) The triangle is obtuse – angled
 - (b) The triangle is acute –angled but not right –angled
 - (c) The triangle is right – angled
 - (d) The triangle is isosceles but not obtuse – angled
7. If median of the $\triangle ABC$ through A is perpendicular to BC, then which one of the following is correct?
 - (a) $\tan A + \tan B = 0$
 - (b) $\tan B - \tan C = 0$
 - (c) $\tan C + 2 \tan A = 0$
 - (d) $\tan B + \tan C = 0$
8. In a triangle ABC, $b = \sqrt{3}$ cm, $c = 1$ cm, $\angle A = 30^\circ$, what is the value of a?
 - (a) $\sqrt{2}$
 - (b) 2 cm
 - (c) 1 cm
 - (d) $\frac{1}{2}$ cm
9. ABC is a triangle in which $AB = 6$ m, $BC = 8$ cm and $CA = 10$ cm. What is the value of $\cot(A/4)$?
 - (a) $\sqrt{5} - 2$
 - (b) $\sqrt{5} + 2$
 - (c) $\sqrt{3} - 1$
 - (d) $\sqrt{3} + 1$
10. If the sides of a triangle are 6cm, 10cm and 14cm, then what is the largest angle included by the sides?
 - (a) 90°
 - (b) 120°
 - (c) 135°
 - (d) 150°
11. In a triangle ABC, $AC = 5$ cm, $AB = 7$ cm, $BC = \sqrt{3}$. What is the measure of angle A?
 - (a) 45°
 - (b) 60°
 - (c) 90°
 - (d) 300°
12. ABC is a triangle in which $BC = 10$ cm, $CA = 6$ cm and $AB = 8$ cm, which one of the following is a correct?
 - (a) ABC is an acute angled triangle
 - (b) ABC is an obtuse angled triangle
 - (c) ABC is an right angled triangle
 - (d) None of the above

ANSWER KEY

1.	c	2.	d	3.	a	4.	b	5.	c	6.	c	7.	b	8.	b	9.	b	10.	b
11.	b	12.	c																

Solutions

Sol. 1. (c)

From properties of triangle we know that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Given that $a = 2b$ and $A = 3B$.

$$\text{We get } \frac{2b}{\sin 3B} = \frac{b}{\sin B}$$

$$\Rightarrow \frac{\sin 3B}{\sin B} = 2 \Rightarrow \frac{3\sin B - 4\sin^3 B}{\sin B} = 2$$

$$\Rightarrow 3 - 4\sin^2 B = 2$$

$$\Rightarrow \sin^2 B = \left(\frac{1}{2}\right)^2 \Rightarrow \sin B = \frac{1}{2} = \sin \frac{\pi}{6}$$

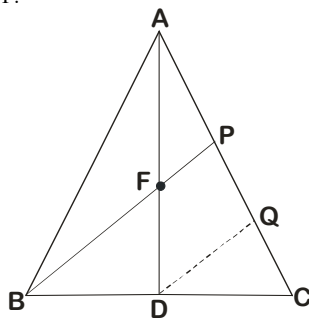
$$\Rightarrow B = \frac{\pi}{6} \text{ since } A = 3B$$

$$\text{So, } \angle A = \frac{\pi}{2}$$

Thus, triangle is right angled triangle.

Sol. 2. (d)

Given, AD is the median of $\triangle ABC$. F is mid-point of AD and BF is produced to meet the side AC in P.



Draw $DQ \parallel BP$

In $\triangle ADQ$, F is mid-point of AD and $FP \parallel DQ$.

$\therefore$ P is mid-point of AQ (converse of mid-point theorem)

$$\Rightarrow AP = PQ$$

In $\triangle BCP$, D is mid-point of BC and $DQ \parallel BP$.

$\therefore$ Q is midpoint of PC

$$\Rightarrow PQ = QC$$

From (1), (2) we get, $AP = PQ = QC$

From figure, $AP + PQ + QC = AC$

$$\Rightarrow AP + AP + AP = AC$$

$$\Rightarrow 3 \times AP = AC$$

$$\Rightarrow AP = \frac{1}{3} \times AC \quad \therefore \lambda = \frac{1}{3}$$

Sol. 3. (a)

We know from properties of triangle that

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}} \Rightarrow \cos^2 \frac{A}{2} = \frac{s(s-a)}{bc}$$

$$\text{and } \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}} \Rightarrow \cos^2 \frac{C}{2} = \frac{s(s-c)}{ab}$$

$$\text{So, } a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2}$$

$$= a \left(\frac{s(s-c)}{ab} \right) + c \left(\frac{s(s-a)}{bc} \right) = \frac{s(s-c+s-a)}{b}$$

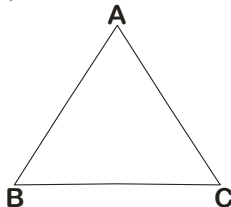
$$= \frac{s(2s-a-c)}{b} = \frac{s(a+b+c-a-c)}{b} = \frac{s \cdot b}{b} = s$$

$$= \frac{30}{2} = 15 \text{ cm} \quad [\text{given that } 2s = 30]$$

Sol. 4. (b)

Ratio of angles is given by $\angle A : \angle B : \angle C = 1 : 2 : 3$

Let $\angle A = x$, $\angle B = 2x$ and $\angle C = 3x$



We know that in a triangle

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow x + 2x + 3x = 180^\circ$$

$$\Rightarrow x = \frac{180^\circ}{6} = 30^\circ$$

So, $\angle A = 30^\circ$, $\angle B = 60^\circ$ and $\angle C = 90^\circ$

From sin law

$$\frac{BC}{\sin A} = \frac{CA}{\sin B} = \frac{AB}{\sin C} = K$$

$$\frac{BC}{\sin 30^\circ} = \frac{CA}{\sin 60^\circ} = \frac{AB}{\sin 90^\circ} = K$$

$$BC = K \sin 30^\circ$$

$$CA = K \sin 60^\circ$$

$$AB = K \sin 90^\circ$$

$$BC : CA : AB$$

$$= \sin 30^\circ : \sin 60^\circ : \sin 90^\circ$$

$$= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 = 1 : \sqrt{3} : 2$$

Sol. 5. (c)

Let the angles A, B and C of a triangle are $2x$, $5x$ and $5x$, respectively

So, $2x + 5x + 5x = 180^\circ$

$$\Rightarrow x = \frac{180^\circ}{12} = 15^\circ$$

Angles are 30° , 75° , 75°

$\angle B = 75^\circ$ and $\angle C = 75^\circ$

$$\therefore \tan B \text{ and } C = (\tan 75^\circ)^2 = (\tan(45^\circ + 30^\circ))^2$$

$$= \left(\frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} \right)^2 = \left(\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \right)^2 = \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right)^2$$

$$= \left(\frac{\sqrt{3} + 1}{3 - 1} \right)^2 = \frac{1}{4} [3 + 1 + 2\sqrt{3}]^2$$

$$= \frac{1}{4} [4 + 2\sqrt{3}]^2 = \frac{1}{4} [16 + 12 + 16\sqrt{3}]$$

$$= \frac{1}{4} [28 + 16\sqrt{3}] = 7 + 4\sqrt{3}$$

Sol. 6. (c)

We know from the sine law that

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow \frac{2b}{\sin 3B} = \frac{b}{\sin B}$$

$$\Rightarrow 2 \sin B = \sin 3B$$

$$\Rightarrow 2 \sin B = 3 \sin B - 4 \sin^3 B$$

$$\Rightarrow \sin B - 4 \sin^3 B = 0$$

$$\Rightarrow \sin B (1 - 4 \sin^2 B) = 0$$

$$\Rightarrow \sin B = 0 \text{ or } 1 - 4 \sin^2 B = 0 \Rightarrow B = 0 \text{ or } B = 30^\circ$$

$$\Rightarrow B = 30^\circ \text{ and } A = 3 \times 30^\circ = 90^\circ$$

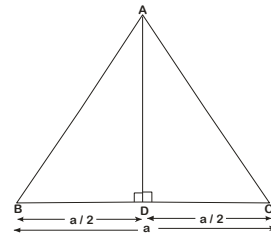
$$\Rightarrow B = 0 \text{ is not possible so, } B = 30^\circ \text{ and } A = 3 \times 30^\circ = 90^\circ$$

$\Rightarrow$ The triangle is right angled triangle.

Sol. 7. (b)

In the given $\triangle ABC$

Let $BC = a$



$$\therefore BD = CD = \frac{a}{2}$$

In $\triangle ADB$,

$$\tan B = \frac{AD}{BD} = \frac{AD}{a/2}$$

$$\Rightarrow \tan B = \frac{2AD}{a} \quad \dots(1)$$

In $\triangle ADC$

$$\tan C = \frac{AD}{CD} = \frac{AD}{a/2}$$

$$\tan C = \frac{2AD}{a} \quad \dots(2)$$

From eqs.(1) and (2), we get $\tan B = \tan C$

$$\Rightarrow \tan B - \tan C = 0$$

Sol. 8. (c)

As given : In a triangle ABC, $AC = b = \sqrt{3}$ cm,

$AB = c = 1$ m and $\angle A = 30^\circ$

From cosine formulae

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{(\sqrt{3})^2 + 1^2 - a^2}{2\sqrt{3} \cdot 1}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{3 + 1 - a^2}{2\sqrt{3}} \Rightarrow 3 = 4 - a^2$$

$$\Rightarrow a^2 = 4 - 3 = 1 \Rightarrow a = 1 \text{ cm}$$

Sol. 9. (b)

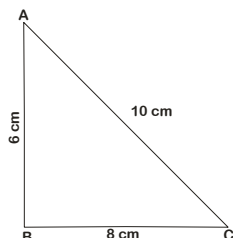
Here, AB = 6 cm, BC = 8 cm and CA = 10 cm

$$S = \frac{a + b + c}{2} = \frac{24}{2} = 12$$

$$\therefore \tan \frac{A}{2} = \sqrt{\frac{(12-10)(12-6)}{12(12-8)}}$$

$$\left(\because \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \right)$$

$$= \sqrt{\frac{1}{4}} = \frac{1}{2}$$



$$\therefore \cot A/2 = 2$$

$$\text{Now, } \cot \left(\frac{A}{4} + \frac{A}{4} \right) = \frac{\cot^2 \frac{A}{4} - 1}{2 \cot \frac{A}{4}}$$

$$\cot \left(\frac{A}{2} \right) = \frac{\cot^2 \frac{A}{4} - 1}{2 \cot \frac{A}{4}}$$

$$\text{Let } \cot \left(\frac{A}{4} \right) = x$$

$$\therefore 2 = \frac{x^2 - 1}{2x}$$

$$\Rightarrow x^2 - 4x - 1 = 0$$

$$\Rightarrow x = \frac{4 \pm \sqrt{16 + 4}}{2}$$

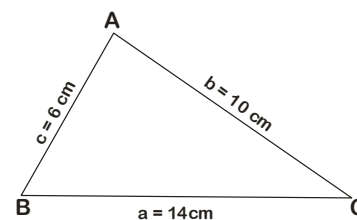
$$\Rightarrow x = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$$

$$\text{So, } \cot \left(\frac{A}{4} \right) = \sqrt{5} + 2 \text{ or } 2 - \sqrt{5}$$

Sol. 10. (b)

We know that largest side has greatest angle opposite it,

$$\therefore a = 14 \text{ cm, } b = 10 \text{ cm and } c = 6 \text{ cm}$$



$$\therefore \cos A = \frac{c^2 + b^2 - a^2}{2bc}$$

$$= \frac{36 + 100 - 96}{2 \times 6 \times 10} = -\frac{1}{2} = \cos 120^\circ$$

$$\Rightarrow \angle A = 120^\circ$$

Sol. 11. (b)

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{25 + 49 - 39}{2 \times 5 \times 7}$$

$$\cos A = 1/2$$

$$A = 60^\circ$$

Sol. 12. (c)

$$6^2 + 8^2 = 10^2$$

so triangle is right angles

NDA PYQ

Direction (For next four):

ABC is a triangle right angled at B. The hypotenuse (AC) is four times the perpendicular (BD) drawn to it from the opposite vertex and $AD < DC$.

1. What is one of the acute angles of the triangle?

(a) 15° (b) 30°
(c) 45° (d) none

[NDA (I) - 2011]

2. What is $\angle ABD$?

(a) 15° (b) 30°
(c) 45° (d) none

[NDA (I) - 2011]

3. What is AD : DC equal to?

(a) $(7 - 2\sqrt{3}) : 1$ (b) $(7 - 4\sqrt{3}) : 1$
(c) 1 : 2 (d) none of these

[NDA (I) - 2011]

4. What is $\tan(A - C)$ equal to?

(a) 0 (b) 1
(c) 2 (d) none

[NDA (I) - 2011]

5. In a triangle ABC, if $c = 2$, $A = 120^\circ$ and $a = \sqrt{6}$, then what is angle C equal to

(a) 75° (b) 60°
(c) 45° (d) 30°

[NDA (I) - 2011]

6. If the angles of a triangle ABC are in AP and $b : c = \sqrt{3} : \sqrt{2}$, then what is the measure of angle A?

(a) 30° (b) 45°
(c) 60° (d) 75°

[NDA (I) - 2011]

7. The sides a, b, c of a $\triangle ABC$ are in AP and 'a' is the smallest side. What is $\cos A$ equal to?

(a) $\frac{3c - 4b}{2c}$ (b) $\frac{3c - 4b}{2b}$
(c) $\frac{4c - 3b}{2c}$ (d) $\frac{3b - 4c}{2c}$

[NDA (II) - 2011]

8. In a $\triangle ABC$, $a = 8$, $b = 10$ and $c = 12$. What is $\angle C$ equal?

(a) $A/2$ (b) $2A$
(c) $3A$ (d) $3A/2$

[NDA (II) - 2011]

9. If sides of a triangle are in the ratio $2 : \sqrt{6} : 1 + \sqrt{3}$, then what is the smallest angle of the triangle?

(a) 75° (b) 60°
(c) 45° (d) 30°

[NDA (II) - 2011]

10. If the sides of a triangle are 6cm, 10cm and 14cm. Then the triangle is obtuse angled with the obtuse angle equal to

(a) 150° (b) 135°
(c) 120° (d) 105°

[NDA (II) - 2011]

11. In a $\triangle ABC$, if the angles A, B, C are in AP, then which one of the following is correct?

(a) $c = a + b$ (b) $c^2 = a^2 + b^2 - ab$
(c) $a^2 = b^2 + c^2 - bc$ (d) $b^2 = a^2 + c^2 - ac$

[NDA (I) - 2012]

12. In any $\triangle ABC$, $a = 18$, $b = 24$ and $c = 30$, then what is $\sin C$ equal?

(a) $1/4$ (b) $1/3$
(c) $1/2$ (d) 1

[NDA (I) - 2013]

13. If the angles of a triangle are 30° , 45° and the included side is $\sqrt{3} + 1$, then what is the area of triangle?

(a) $\left(\frac{\sqrt{3}+1}{2}\right)cm^2$ (b) $(2\sqrt{3}+1)cm^2$

(c) $\left(\frac{\sqrt{3}+1}{3}\right)cm^2$ (d) $\left(\frac{\sqrt{3}-1}{2}\right)cm^2$

[NDA (II) - 2013]

14. Consider the following statements:

I. There exists no $\triangle ABC$ for which $\sin A + \sin B = \sin C$
II. If the angles of a triangle are in the ratio 1 : 2 : 3, then its sides will be in the ratio 1 : $\sqrt{3}$: 2

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

15. In a $\triangle ABC$, if $c = 2$, $A = 45^\circ$ and $a = 2\sqrt{2}$, then what is C equal to?

(a) 30° (b) 15°
(c) 45° (d) none

[NDA (II) - 2014]

16. In a triangle ABC, $\sin A - \cos B = \cos C$, then what is B equal to?

(a) π (b) $\pi/3$
(c) $\pi/2$ (d) $\pi/4$

[NDA-2014(2)]

17. In a $\triangle ABC$, $a = (1 + \sqrt{3})$, $b = 2$ cm, and $\angle C = 60^\circ$, then the other two angles are

(a) 45° and 75° (b) 30° and 90°
(c) 105° and 15° (d) 100° and 20°

[NDA (I) - 2015]

Direction (for next two): Consider a $\triangle ABC$ satisfying

$$2a \sin^2\left(\frac{C}{2}\right) + 2c \sin^2\left(\frac{A}{2}\right) = 2a + 2c - 3b$$

18. The sides of the triangle are in

(a) GP (b) AP
(c) HP (d) None

[NDA (II) - 2015]

19. $\sin A$, $\sin B$, $\sin C$ are in

(a) GP (b) AP
(c) HP (d) None

[NDA (II) - 2015]

20. If a, b, c are the sides of a triangle ABC, then $\frac{1}{a^p} + \frac{1}{b^p} - \frac{1}{c^p}$ where $p > 1$, is :

(a) always negative
(b) always positive
(c) always zero
(d) positive if $1 < p < 2$ and negative if $p > 2$

[NDA-2015(2)]

21. Consider the following statements:

1. If ABC is an equilateral triangle, then $3 \tan(A+B) \tan C = 1$
2. If ABC is a triangle in which $A = 78^\circ$, $B = 66^\circ$, then

$$\tan \left(\frac{A}{2} + C \right) < \tan A$$

$$3. \text{ If } ABC \text{ is any triangle, then } \tan \left(\frac{A+B}{2} \right) \sin \left(\frac{C}{2} \right) < \cos \left(\frac{C}{2} \right)$$

Which of the above statements is/are correct?

- (a) 1 only (b) 2 only
(c) 1 and 2 (d) 2 and 3

[NDA-2016(1)]

Direction (for next two): Consider a ΔABC in which

$$\cos A + \cos B + \cos C = \sqrt{3} \sin \frac{\pi}{3}$$

$$22. \text{ What is the value of } \sin \left(\frac{A}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{C}{2} \right)$$

- (a) 1/2 (b) 1/4
(c) 1/8 (d) 1/8

[NDA (I) - 2016]

$$23. \text{ What is the value of } \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{B+C}{2} \right) \cos \left(\frac{C+A}{2} \right)$$

- (a) 1/4 (b) 1/2
(c) 1/16 (d) None

[NDA (I) - 2016]

24. Consider the following for triangle ABC

$$\text{I. } \sin \left(\frac{B+C}{2} \right) = \cos \left(\frac{A}{2} \right)$$

$$\text{II. } \tan \left(\frac{B+C}{2} \right) = \cot \left(\frac{A}{2} \right)$$

$$\text{III. } \sin(B+C) = \cos A$$

$$\text{IV. } \tan(B+C) = -\cot A$$

Which of the above are correct ?

- (a) I and III (b) I and II
(c) I and IV (d) II and III

[NDA (I) - 2017]

$$25. \text{ In a } \Delta ABC, a - 2b + c = 0, \text{ then the value of } \cot \left(\frac{A}{2} \right) \cot \left(\frac{C}{2} \right)$$

- (a) 9/2 (b) 3
(c) 3/2 (d) 1

[NDA (II) - 2017]

$$26. \text{ In a } \Delta ABC, \text{ If } \frac{\sin^2 A + \sin^2 B + \sin^2 C}{\cos^2 A + \cos^2 B + \cos^2 C} = 2 \text{ then the triangle is}$$

- (a) right angled (b) equilateral
(c) isosceles (d) obtuse angled

[NDA (II) - 2017]

27. If x , $x-y$ and $x+y$ are the angles of a triangle (not an equilateral triangle) such that $\tan(x-y)$, $\tan x$ and $\tan(x+y)$ are in GP, then what is x equal to?

- (a) $\pi/4$ (b) $\pi/3$
(c) $\pi/6$ (d) $\pi/2$

[NDA-2018(1)]

28. In a ΔABC , If $a = 2$, $b = 3$, and $\sin A = 2/3$ then what is value of angle B

- (a) 45° (b) 90°
(c) 60° (d) 30°

[NDA (I) - 2018]

29. ABC is a triangle inscribed in a circle with centre O, Let $\alpha = \angle BAC$, where $45^\circ < \alpha < 90^\circ$, let $\beta = \angle BOC$, which one of the following is correct?

- (a) $\cos \beta = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$ (b) $\cos \beta = \frac{1 + \tan^2 \alpha}{1 - \tan^2 \alpha}$
(c) $\cos \beta = \frac{2 \tan \alpha}{1 + \tan^2 \alpha}$ (d) $\sin \beta = 2 \sin^2 \alpha$

[NDA (I) - 2018]

30. If $(A + B + C) = 180^\circ$, then what is $\sin 2A - \sin 2B - \sin 2C =$

- (a) $-4 \sin A \cdot \sin B \cdot \sin C$ (b) $-4 \cos A \cdot \sin B \cdot \cos C$
(c) $-4 \cos A \cdot \cos B \cdot \sin C$ (d) $-4 \sin A \cdot \cos B \cdot \cos C$

[NDA (II) - 2018]

31. If the angles of a triangle ABC are in the ratio 1:2:3, then the corresponding sides are in the ratio?

- (a) 1 : 2 : 3 (b) 3 : 2 : 1
(c) 1 : $\sqrt{3}$: 2 (d) 1 : $\sqrt{3}$: $\sqrt{2}$

[NDA (I) - 2019]

32. If angle C of a triangle ABC is a right angle, then what is $\tan A + \tan B$ equal to?

- (a) $\frac{a^2 - b^2}{ab}$ (b) $\frac{a^2}{bc}$
(c) $\frac{b^2}{ca}$ (d) $\frac{c^2}{ab}$

[NDA-2019(2)]

33. If the angles of a triangle ABC are in AP and $b : c = \sqrt{3} : \sqrt{2}$, then what is the measure of angle A?

- (a) 30° (b) 45°
(c) 60° (d) 75°

[NDA (II) - 2019]

34. Consider the following statements :

1. If in a triangle ABC, $A = 2B$ and $b = c$, then it must be an obtuse angled triangle.

2. There exists no triangle ABC with $A = 40^\circ$, $B = 65^\circ$ and

$$\frac{a}{c} = \sin 40^\circ \csc 15^\circ$$

Select the correct answer using the code given below

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

35. Consider the following statements :

1. If ABC is a right angled triangle, right angled at A and if $\sin B = 1/3$, then $\csc C = 3$

2. If $b \cos B = c \cos C$ and if the triangle ABC is not a right angled triangle then ABC must be isosceles

Select the correct answer using the code given below

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

Direction (For next two):

Read the following information and answer the given items
ABCD is a trapezium such that AB and CD are parallel and BC is perpendicular to them, Let $\angle ABD = \alpha$, $\angle ADB = \theta$, $BC = p$ and $CD = q$

36. Consider the following:

1. $AD \sin \theta = AB \sin \alpha$
2. $BD \sin \theta = AB \sin(\theta + \alpha)$

Select the correct answer using the code given below

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

37. What AB equal to?

- (a) $\frac{(p^2 + q^2) \sin \theta}{p \cos \theta + q \sin \theta}$ (b) $\frac{(p^2 - q^2) \sin \theta}{p \cos \theta + q \sin \theta}$
 (c) $\frac{(p^2 + q^2) \sin \theta}{q \cos \theta + p \sin \theta}$ (d) $\frac{(p^2 - q^2) \sin \theta}{q \cos \theta + p \sin \theta}$

[NDA 2020]

38. A chord subtends an angle 120° at the centre of a unit circle. What is the length of the chord?

- (a) $\sqrt{2} - 1$ units (b) $\sqrt{3} - 1$ units
 (c) $\sqrt{2}$ units (d) $\sqrt{3}$ units

[NDA 2021 (I)]

39. The sides of a triangle are m, n and $\sqrt{m^2 + n^2 + mn}$. What is the sum of the acute angles of the triangle?

- (a) 45° (b) 60°
 (c) 75° (d) 90°

[NDA 2021 (I)]

40. What is the area of the triangle ABC with sides $a = 10$ cm, $c = 4$ cm and angle $B = 30^\circ$?

- (a) 16 cm^2 (b) 12 cm^2
 (c) 10 cm^2 (d) 8 cm^2

[NDA 2021 (I)]

41. Let ABC be a triangle. If $\cos 2A + \cos 2B + \cos 2C = -1$, then which one of the following is correct?

- (a) $\sin A \sin B \sin C = 0$ (b) $\sin A \sin B \cos C = 0$
 (c) $\cos A \sin B \sin C = 0$ (d) $\cos A \cos B \cos C = 0$

[NDA (II) - 2021]

42. In a triangle ABC, $\sin A - \cos B - \cos C = 0$. What is single B equal to?

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

[NDA 2022 (I)]

43. Let a, b, c be the lengths of sides BC, CA, AB respectively of a triangle ABC. If p is the perimeter and q is the area of the triangle, then what is $p(p-2a)\tan\left(\frac{A}{2}\right)$ equal to?

- (a) q (b) 2q
 (c) 3q (d) 4q

[NDA 2022 (I)]

Consider the following for the next three (03) items that follow:

ABC is a triangular plot with $AB = 16\text{m}$, $BC = 10\text{m}$ and $CA = 10\text{m}$. A lamp post is situated at the middle point of the side AB. The lamp post subtends an angle 45° at the vertex B.

44. What is the height of the lamp post?

- (a) 6m (b) 7m
 (c) 8m (d) 9m

[NDA 2022 (II)]

45. What is $\frac{AB}{\sin C}$ equal to?

- (a) 17m (b) $\frac{50}{3}$ m
 (c) $\frac{40}{3}$ m (d) 16m

[NDA 2022 (II)]

46. What is $\cos A + \cos B + \cos C$ equal to?

- (a) 1 (b) $\frac{41}{25}$

- (c) $\frac{37}{25}$ (d) $\frac{33}{25}$

[NDA 2022 (II)]

47. In a triangle ABC, $a = 4$, $b = 3$, $c = 2$. What is $\cos 3C$ equal to?

- (a) $\frac{7}{128}$ (b) $\frac{11}{128}$
 (c) $\frac{7}{64}$ (d) $\frac{11}{64}$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$.

Further the in-circle of the triangle touches the sides PQ, QR and RP at N, L and M respectively such that the length PN, QL and RM are n, $n + 2$, $n + 4$ respectively where n is an integer.

48. What is the value of n?

- (a) 4 (b) 6
 (c) 8 (d) 10

[NDA - 2023 (1)]

49. What is the length of the smallest side?

- (a) 12 (b) 14
 (c) 16 (d) 18

[NDA - 2023 (1)]

Consider the following for the next two (02) items that follow:

The perimeter of a triangle ABC is 6 times the AM of sine of angles of the triangle. Further $BC = \sqrt{3}$ and $CA = 1$.

50. What is the perimeter of the triangle?

- (a) $\sqrt{3} + 1$ (b) $\sqrt{3} + 2$
 (c) $\sqrt{3} + 3$ (d) $2\sqrt{3} + 1$

[NDA - 2023 (1)]

51. Consider the following statements:

1. ABC is right angled triangle

2. The angles of the triangle are in AP

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
 (c) both 1 and 2 (d) neither 1 nor 2

[NDA - 2023 (1)]

Consider the following for the next two (02) items that follow:

In the triangle ABC, $a^2 + b^2 + c^2 = ac + \sqrt{3} bc$

52. What is the nature of the triangle?

- (a) equilateral
 (b) isosceles
 (c) right angled triangle
 (d) scalene but not right angles

[NDA - 2023 (1)]

53. If $c = 8$, what is the area of the triangle?

- (a) $4\sqrt{3}$ (b) $6\sqrt{3}$
 (c) $8\sqrt{3}$ (d) $12\sqrt{3}$

[NDA - 2023 (1)]

Consider the following for the next (02) items

The angles A, B and C of a triangle ABC are in the ratio 3:5:4.

54. What is the value of $a + b + \sqrt{2}c$ equal to?

- (a) 3a (b) 2b
 (c) 3b (d) 2c

[NDA-2023 (2)]

55. What is the ratio of $a^2 : b^2 : c^2$?

- (a) $2 : 2 + \sqrt{3} : 3$ (b) $2 : 2 - \sqrt{3} : 2$

- (c) $2 : 2 + \sqrt{3} : 2$ (d) $2 : 2 - \sqrt{3} : 3$ [NDA-2023 (2)]
56. In a triangle ABC, AB = 16 cm, BC = 63 cm and AC = 65 cm. What is the value of $\cos 2A + \cos 2B + \cos 2C$?
 (a) -1 (b) 0
 (c) 1 (d) $\frac{76}{65}$ [NDA-2024 (1)]
57. Consider the following statements
 1. In a triangle ABC, if $\cot A \cdot \cot B \cdot \cot C > 0$, then the triangle is an acute angled triangle
 2. In a triangle ABC, if $\tan A \cdot \tan B \cdot \tan C > 0$, then the triangle is an obtuse angled triangle
 Which of the statements given above are correct?
 (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2 [NDA-2024 (1)]
58. ABC is a triangle such that angle C = 60° , then what is $\frac{\cos A + \cos B}{\cos\left(\frac{A+B}{2}\right)}$ equal to?
 (a) 2 (b) $\sqrt{2}$
 (c) 1 (d) $\frac{1}{\sqrt{2}}$ [NDA-2024 (1)]
59. In a triangle ABC $\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$ what is the area of the triangle if a = 6 cm?
 (a) $9\sqrt{3}$ square cm (b) 12 square cm
 (c) $18\sqrt{3}$ square cm (d) 24 square cm [NDA-2024 (2)]
60. The roots of the equation $7x^2 - 6x + 1 = 0$ are $\tan \alpha$ and $\tan \beta$, where 2α and 2β are the angles of a triangle. Which one of the following is correct?
 (a) The triangle is equilateral
 (b) The triangle is isosceles but not right-angled
 (c) The triangle is right-angled
 (d) The triangle is right-angled isosceles [NDA-2024 (2)]
61. In a triangle ABC, $\angle A = 75^\circ$ and $\angle B = 45^\circ$. What is $2a - b$ equal to?
 (a) c (b) $\sqrt{2}c$ (c) $2c$ (d) $2\sqrt{2}c$ [NDA-2024 (2)]
62. In a triangle ABC $\tan A + \tan B + \tan C = k$. What is the value of $\cot A \cot B \cot C$?
 (a) $0.5k$ (b) $1/k$
 (c) $3/k$ (d) $1/k^3$ [NDA-2024 (2)]
- Consider the following for the three (03) items that follow:
 The sides of a triangle are AB = 3 cm, BC = 5 cm, CA = 7 cm
63. Consider the following statements:
 I. The triangle is obtuse-angled triangle.
 II. The sum of acute angles of the triangle is also acute.
 Which of the statements given above is/are correct?
 (a) I only (b) II only
 (c) Both I and II (d) neither I nor II [NDA-2025 (1)]
64. What is $\angle B$ equal to?
 (a) 60° (b) 105°
 (c) 120° (d) 150° [NDA-2025 (1)]
65. What is the area of the triangle?
 (a) $15\sqrt{3}/4$ sq. cm (b) $15\sqrt{3}/2$ sq. cm
 (c) $15\sqrt{3}$ sq. cm (d) $30\sqrt{3}$ sq. cm [NDA-2025 (1)]
- Consider the following for the two (02) items that follow:
 In a triangle ABC, two sides BC and CA are in the ratio 2:1 and their opposite corresponding angles are in the ratio 3:1.
66. One of the angles of the triangle is
 (a) 15° (b) 30°
 (c) 45° (d) 75° [NDA-2025 (1)]
67. Consider the following statements
 I. The triangle is right-angled.
 II. One of the sides of the triangle is 3 times the other.
 III. The angles A, C and B of the triangle are in A.P.
 Which of the statements given above is/are correct?
 (a) I only (b) II and III only
 (c) I and III only (d) I, II and III [NDA-2025 (1)]

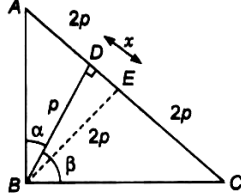
ANSWER KEY

1.	a	2.	a	3.	b	4.	d	5.	c	6.	d	7.	c	8.	b	9.	c	10.	c
11.	d	12.	d	13.	a	14.	c	15.	a	16.	c	17.	a	18.	b	19.	b	20.	a
21.	b	22.	c	23.	d	24.	b	25.	b	26.	a	27.	b	28.	b	29.	a	30.	d
31.	c	32.	d	33.	d	34.	b	35.	d	36.	c	37.	a	38.	d	39.	b	40.	c
41.	d	42.	d	43.	d	44.	c	45.	b	46.	d	47.	a	48.	c	49.	d	50.	c
51.	c	52.	c	53.	c	54.	c	55.	a	56.	a	57.	a	58.	c	59.	a	60.	c
61.	b	62.	b	63.	c	64.	c	65.	a	66.	b	67.	c						

Solutions

Sol.1. (a)

Let $BD = p$ and $DE = x$



$AC = 4p$

Let E be a mid point of line AC.

Then $AE = EC = BE = 2p$

in $\triangle BDE$, $(BE)^2 = (BD)^2 + (ED)^2$

$$(2p)^2 = p^2 + x^2$$

$$x = \sqrt{3}p$$

$$\text{Now } AD = 2p - x = 2p - \sqrt{3}p$$

$$DC = 2p + x = 2p + \sqrt{3}p$$

$$\text{In } \triangle BAD, \tan A = \frac{BD}{AD} = \frac{p}{2p - p\sqrt{3}}$$

$$\tan A = \frac{1}{2 - \sqrt{3}} = 2 + \sqrt{3}$$

$A = 75^\circ$ then $C = 15^\circ$

then $\alpha = 15^\circ$ and $\beta = 75^\circ$

Sol.2. (a)

$\angle ABD = 15^\circ$

Sol.3. (b)

$$AD : DC = (7 - 4\sqrt{3}) : 1$$

Sol.4. (d)

$$\tan(A - C) = \tan(75^\circ - 15^\circ) = \tan 60^\circ = \sqrt{3}$$

Sol.5. (c)

by sine formula

$$\frac{\sin A}{a} = \frac{\sin C}{c} \Rightarrow \frac{\sin 120^\circ}{\sqrt{6}} = \frac{\sin C}{2}$$

by solving this $C = 45^\circ$

Sol.6. (i)

Angles A, B, C are in AP then angle $B = 60^\circ$

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 60^\circ}{\sqrt{3}} = \frac{\sin C}{\sqrt{2}}$$

By solving it $C = 45^\circ$ then $A = 75^\circ$

Sol.7. (c)

a, b, c are in AP

$$2b = a + c$$

$$\cos A = \frac{b^2 + c^2 - (2b - c)^2}{2bc} = \frac{b^2 + c^2 - 4b^2 - c^2 + 4bc}{2bc}$$

$$\cos A = \frac{4bc - 3b^2}{2bc} = \frac{4c - 3b}{2c}$$

Sol.8. (b)

$a = 8$ cm, $b = 10$ cm, $c = 12$ cm

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos C = \frac{8^2 + 10^2 - 12^2}{2 \cdot 8 \cdot 10} = \frac{1}{8}$$

now

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{10^2 + 12^2 - 8^2}{2 \cdot 10 \cdot 12} = \frac{3}{4}$$

$$\cos 2A = 2\cos^2 A - 1 = 2\left(\frac{3}{4}\right)^2 - 1 = \frac{1}{8}$$

so $C = 2A$

Sol.9. (c)

Given $a : b : c = 2 : \sqrt{6} : (\sqrt{3} + 1)$

a is minimum so angle A will be smallest

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{6 + (\sqrt{3} + 1)^2 - 4}{2(\sqrt{6})(\sqrt{3} + 1)}$$

$$\cos A = \frac{6 + 4 + 2\sqrt{3} - 4}{2(\sqrt{6})(\sqrt{3} + 1)} = \frac{6 + 2\sqrt{3}}{2(\sqrt{6})(\sqrt{3} + 1)}$$

$$\cos A = \frac{2\sqrt{3}(\sqrt{3} + 1)}{2(\sqrt{6})(\sqrt{3} + 1)} = \frac{1}{\sqrt{2}}$$

$A = 45^\circ$

Sol.10. (c)

angle opposite to largest side is always largest in a triangle.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos C = \frac{6^2 + 10^2 - 14^2}{2 \cdot 6 \cdot 14}$$

$$\cos C = \frac{-1}{2} \Rightarrow C = 120^\circ$$

Sol.11. (d)

if angles are in AP then $2B = A + C$

and for triangle $A + B + C = 180^\circ$

by above two $\angle B = 60^\circ$

now cosine formula

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\frac{1}{2} = \frac{a^2 + c^2 - b^2}{2ac}$$

$$a^2 + c^2 - b^2 = ac$$

$$b^2 = a^2 + c^2 - ac$$

Sol.12. (d)

we know that $18^2 + 24^2 = 30^2$

so triangle is right angled.

and $\sin 90^\circ = 1$

Sol.13. (a)

$A = 30^\circ$, $B = 45^\circ$ and $c = \sqrt{3} + 1$

then Angle $C = 105^\circ$

now by sine formula

$$\frac{\sin A}{a} = \frac{\sin C}{c} \Rightarrow \frac{\sin 30^\circ}{a} = \frac{\sin 105^\circ}{\sqrt{3} + 1}$$

$$\Rightarrow \frac{1/2}{a} = \frac{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\sqrt{3}+1} \Rightarrow a = \sqrt{2}$$

$$\text{Area} = \frac{1}{2} \times ac \sin B = \frac{1}{2} \times \sqrt{2}(\sqrt{3}+1) \frac{1}{\sqrt{2}}$$

$$\text{Area} = \frac{(\sqrt{3}+1)}{2}$$

Sol.14. (c)

$$\text{I. } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$$

$$\sin A + \sin B = \sin C$$

$$ak + bk = ck$$

$$a + b = c$$

it is not possible for a triangle.

II. $A : B : C = 1 : 2 : 3$

then $A = 30^\circ$, $B = 60^\circ$, $C = 90^\circ$

for these angles

$$a : b : c = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 = 1 : \sqrt{3} : 2$$

Sol.15. (a)

by sine rule

$$\frac{\sin A}{a} = \frac{\sin C}{c} \Rightarrow \frac{\sin 45^\circ}{2\sqrt{2}} = \frac{\sin C}{2}$$

$C = 30^\circ$

Sol.16. (c)

$$\sin A - \cos B = \cos C$$

$$\sin A = \cos B + \cos C$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = 2 \cos \frac{B+C}{2} \cos \frac{B-C}{2}$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = 2 \sin \frac{A}{2} \cos \frac{B-C}{2}$$

$$\cos \frac{A}{2} = \cos \frac{B-C}{2}$$

$$\frac{A}{2} = \frac{B-C}{2}$$

$$A = B - C$$

$$A + C = B$$

$$\text{But } A + B + C = 180$$

$$\text{So } B = 90$$

Sol.17. (a)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\Rightarrow \cos 60^\circ = \frac{(1+\sqrt{3})^2 + (2)^2 - c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow \frac{1}{2} = \frac{(1+\sqrt{3})^2 + (2)^2 - c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow \frac{1}{2} = \frac{(1+\sqrt{3})^2 + (2)^2 - c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow 2 + 2\sqrt{3} = 4 + 2\sqrt{3} + 4 - c^2$$

$$\Rightarrow c = \sqrt{6}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{4+6-(1+\sqrt{3})^2}{2 \cdot 2 \cdot \sqrt{3}}$$

$$\cos A = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$$

$A = 75^\circ$, $C = 60^\circ$ then $B = 45^\circ$

Sol.18. (b)

$$2a \sin^2 \left(\frac{C}{2} \right) + 2c \sin^2 \left(\frac{A}{2} \right) = 2a + 2c - 3b$$

$$\Rightarrow a(1 - \cos C) + c(1 - \cos A) = 2a + 2c - 3b$$

$$\Rightarrow (a+c) - (a \cos C + c \cos A) = 2a + 2c - 3b$$

$$\Rightarrow a + c - b = 2a + 2c - 3b$$

$$\Rightarrow a + c = 2b$$

Hence, a, b, c are in A.P.

Sol.19. (b)

Since, a, b, c are AP.

now by Sine formula

$k \sin A, k \sin B, k \sin C$

$\sin A, \sin B, \sin C$ are in AP.

Sol.20. (b)

given that $p > 1$

so take $p = 2$

for triangle $a + b > c$

take $a = 4, b = 9, c = 10$

half power of 4, 9, 10 are 2, 3, 3.2

so $2 + 3 - 3.2$ is positive.

Sol.21. (b)

$3 \tan(A+B) \tan C = 3 \tan 120^\circ \tan 60^\circ$ is negative

statement 1 is wrong.

$\tan(39+66) = \tan 105^\circ$ that is negative

$\tan 78^\circ$ is positive

$\tan 105^\circ < \tan 78^\circ$

statement 2 is correct

solve left hand side than we will get right hand side

so statement 3 is incorrect.

Sol.22. (c)

$$\cos A + \cos B + \cos C = \sqrt{3} \sin \frac{\pi}{3}$$

$$\cos A + \cos B + \cos C = 3/2$$

we know that

$$\cos A + \cos B + \cos C = 1 + 4 \sin \left(\frac{A}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{C}{2} \right)$$

$$\Rightarrow \frac{3}{2} = 1 + 4 \sin \left(\frac{A}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{C}{2} \right)$$

$$\Rightarrow \sin \left(\frac{A}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{C}{2} \right) = \frac{1}{8}$$

Sol.23. (d)

$$= \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{B+C}{2} \right) \cos \left(\frac{C+A}{2} \right)$$

$$= \cos \left(90^\circ - \frac{C}{2} \right) \cos \left(90^\circ - \frac{A}{2} \right) \cos \left(90^\circ - \frac{B}{2} \right)$$

$$= \sin \left(\frac{C}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{A}{2} \right) = \frac{1}{8}$$

$$\Rightarrow \sin \left(\frac{A}{2} \right) \sin \left(\frac{B}{2} \right) \sin \left(\frac{C}{2} \right) = \frac{1}{8}$$

Sol.24. (b)

In triangle ABC, $A + B + C = \pi$

$$\Rightarrow \frac{A+B+C}{2} = \frac{\pi}{2}$$

$$\Rightarrow \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2} - \frac{A}{2}$$

by taking sine both sides

$$\Rightarrow \sin \left(\frac{B+C}{2} \right) = \sin \left(\frac{\pi}{2} - \frac{A}{2} \right) = \cos \frac{A}{2}$$

$$\text{also } \Rightarrow \tan \left(\frac{B+C}{2} \right) = \tan \left(\frac{\pi}{2} - \frac{A}{2} \right) = \cot \frac{A}{2}$$

Sol.25. (b)

$$\cot \left(\frac{A}{2} \right) \cot \left(\frac{C}{2} \right)$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \cdot \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$\Rightarrow \sqrt{\frac{s^2}{(s-b)^2}} = \frac{s}{s-b}$$

$$\frac{\frac{a+b+c}{2}}{\frac{a+b+c}{2} - b} = \frac{2b+b}{2b-b} = 3$$

[given that $2b = a + c$]

Sol.26. (a)

$$\frac{\sin^2 A + \sin^2 B + \sin^2 C}{\cos^2 A + \cos^2 B + \cos^2 C} = 2$$

put $A = 30^\circ$, $B = 60^\circ$ and $C = 90^\circ$

these angles satisfies this equation.

so triangle is right angled.

Sol.27. (b)

sum of angles of a triangle = 180

$$(x-y) + x + (x+y) = 180$$

$$x = 60$$

Sol.28. (b)

by sine formula

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{2/\sqrt{3}}{2} = \frac{\sin B}{3}$$

$B = 90^\circ$

Sol.29. (a)

angle made by a chord on centre is always double to made by a point on circumference.

$$\cos \beta = \cos 2\alpha$$

$$\frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$$

$$\frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$$

Sol.30. (d)

$$\sin 2A - \sin 2B - \sin 2C$$

$$= 2 \cos(A+B) \sin(A-B) + \sin(2A+2B)$$

$$= 2 \cos(A+B) \sin(A-B) + 2 \sin(A+B) \cos(A+B)$$

$$= 2 \cos(A+B) [\sin(A-B) + \sin(A+B)]$$

$$= 2 \cos C [2 \sin A \sin B]$$

$$= 4 \sin A \cos B \cos C$$

Sol.31. (c)

angles are in the ratio 1 : 2 : 3

then angles are $30^\circ, 60^\circ, 90^\circ$

now Sine formula

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{1/2}{a} = \frac{\sqrt{3}/2}{b} = \frac{1}{c}$$

$$a : b : c = 1 : \sqrt{3} : 2 \text{ then}$$

Sol.32. (d)

$$\tan A + \tan B = \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$$

Sol.33. (d)

A, B, C are in AP

$$A + B + C = 180^\circ$$

$$2B = A + C$$

$$\text{so } B = 60^\circ$$

$$\text{now Sine formula } \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin B}{\sin C} = \frac{b}{c}$$

$$\frac{\sqrt{3}/2}{\sin C} = \frac{\sqrt{3}}{\sqrt{2}}$$

$C = 45^\circ$ then Angle $A = 75^\circ$

Sol.34. (d)

statement 1

$b = c$ then $\angle B = \angle C$

$$A + B + C = 180^\circ$$

$$2B + B + B = 180^\circ \Rightarrow B = 45^\circ = C \text{ then } A = 90^\circ$$

triangle is right angled.3

Statement 1 is incorrect.

Statement 2 $A = 40^\circ$, $B = 65^\circ$ then $C = 75^\circ$

$$\text{by sine formula } \frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\Rightarrow \frac{a}{c} = \frac{\sin A}{\sin C} = \frac{\sin 40^\circ}{\sin 75^\circ} = \sin 40^\circ \cos 75^\circ$$

Statement 2 is wrong.

Sol.35. (b)

statement 1

$$A + B + C = 180^\circ$$

$$\sin(A+B) = \sin(180^\circ - C)$$

$$\sin(90^\circ + B) = \sin C$$

$$\cos B = \sin C$$

$$\sec B = \operatorname{cosec} C$$

$$\sin B = 1/3 \text{ (given)}$$

$$\text{then } \sec B = \frac{3}{2\sqrt{2}} = \operatorname{cosec} C$$

Statement 1 is wrong.

Statement 2

$$b \cos B = c \cos C$$

$$b = \frac{c \cos C}{\cos B}$$

$$\text{by sine formula } \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin B}{c \cos C} \cos B = \frac{\sin C}{c}$$

$$\sin B \cos B = \sin C \cos C$$

$$\sin 2B = \sin 2C$$

$$\sin 2B - \sin 2C = 0$$

$$2 \cos(B+C) \sin(B-C) = 0$$

$$\cos(B+C) = 0 \text{ or } \sin(B-C) = 0$$

$$B + C = 90^\circ \text{ or } B = C$$

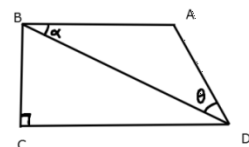
triangle will be right angled or isosceles

But its already given that triangle is not right

angled so triangle is isosceles

Statement 2 is correct.

Sol.36. (c)



Apply Sine formula in $\triangle ABD$

$$\frac{AD}{\sin \alpha} = \frac{AB}{\sin \theta} \Rightarrow AD \sin \theta = AB \sin \alpha$$

statement 1 is correct

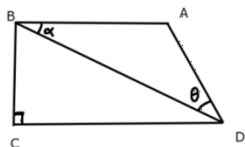
Again apply Sine formula in $\triangle ABD$

$$\frac{BD}{\sin(180 - (\alpha + \theta))} = \frac{AB}{\sin \theta} \Rightarrow \frac{BD}{\sin(\alpha + \theta)} = \frac{AB}{\sin \theta}$$

$$\Rightarrow BD \sin \theta = AB \sin(\alpha + \theta)$$

statement 2 is correct

Sol.37. (a)



given $BC = p$ and $CD = q$
so $BD = \sqrt{p^2 + q^2}$, $\cos \alpha = \frac{q}{\sqrt{p^2 + q^2}}$,

$$\sin \alpha = \frac{p}{\sqrt{p^2 + q^2}}$$

now apply Sine formula in $\triangle ABD$

$$\frac{BD}{\sin(\alpha + \theta)} = \frac{AB}{\sin \theta} \Rightarrow AB = \frac{BD \sin \theta}{\sin(\alpha + \theta)}$$

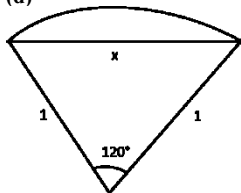
$$\Rightarrow AB = \frac{\sqrt{a^2 + b^2} \sin \theta}{\sin \alpha \cos \theta - \cos \alpha \sin \theta}$$

put all above values

$$\Rightarrow AB = \frac{\sqrt{a^2 + b^2} \sin \theta}{\frac{p}{\sqrt{p^2 + q^2}} \cos \theta + \frac{q}{\sqrt{p^2 + q^2}} \sin \theta}$$

$$\Rightarrow AB = \frac{(p^2 + q^2) \sin \theta}{p \cos \theta + q \sin \theta}$$

Sol.38. (d)

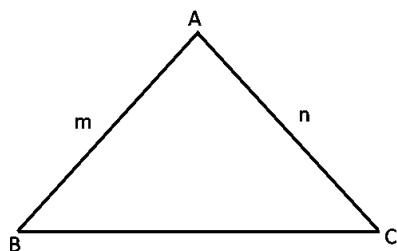


$$\cos \theta = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos 120^\circ = \frac{1 + 1 - x^2}{2} \Rightarrow -\frac{1}{2} = \frac{1 + 1 - x^2}{2}$$

$$\Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3} \text{ unit}$$

Sol.39. (b)



$$\cos A = \frac{m^2 + n^2 - (m^2 + n^2 + mn)}{2mn}$$

$$\cos A = \frac{-mn}{2mn} = -\frac{1}{2}$$

$$A = 120^\circ$$

Sum of other two acute angles will be 60° .

Sol.40. (c)

$$\text{Area of triangle} = \frac{1}{2} ac \sin B$$

$$= \frac{1}{2} \times 10 \times 4 \times \sin 30^\circ$$

$$= \frac{1}{2} \times 10 \times 4 \times \frac{1}{2} = 10 \text{ cm}^2$$

Sol.41. (d)

$$\cos 2A + \cos 2B + \cos 2C = -1$$

$$2 \cos(A+B) \cos(A-B) + \cos 2C = -1$$

$$2 \cos(\pi - C) \cos(A-B) + 2 \cos^2 C - 1 = -1$$

$$-2 \cos C \cos(A-B) + 2 \cos^2 C = 0$$

$$2 \cos C (\cos C - \cos(A-B)) = 0$$

$$2 \cos C \left(-2 \sin \frac{C+A-B}{2} \sin \frac{C-A+B}{2} \right) = 0$$

$$-4 \cos C \sin \left(\frac{\pi - 2B}{2} \right) \sin \left(\frac{\pi - 2A}{2} \right) = 0$$

$$-4 \cos C \cos B \cos A = 0$$

$$\cos A \cos B \cos C = 0$$

Sol.42. (d)

$$\sin A - \cos B - \cos C = 0$$

$$\sin A = \cos B + \cos C$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = 2 \cos \frac{B+C}{2} \cos \left(\frac{B-C}{2} \right)$$

$$\therefore A + B + C = 180$$

$$B + C = 180 - A$$

$$\frac{B+C}{2} = 90 - \frac{A}{2}$$

$$\cos \left(\frac{B+C}{2} \right) = \cos \left(90 - \frac{A}{2} \right) = \sin \frac{A}{2}$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = 2 \sin \frac{A}{2} \cos \frac{B-C}{2}$$

$$\cos \frac{A}{2} = \cos \frac{B-C}{2}$$

$$\frac{A}{2} = \frac{B-C}{2}$$

$$A + C = B$$

$$\text{And } A + B + C = 180^\circ$$

$$\text{So } \angle B = 90^\circ$$

Sol.43. (d)

$$p = \text{perimeter} = 2$$

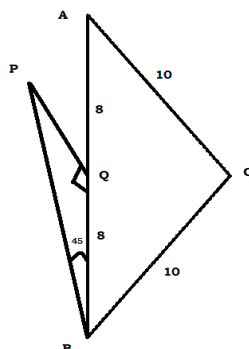
$$2s(2s-2a) \cdot \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$4s(s-a) \cdot \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$= 4 \sqrt{s(s-a)(s-b)(s-c)}$$

$$= 4q$$

Sol.44. (c)



In $\triangle BPQ$

$$\tan 45 = \frac{PQ}{8} \Rightarrow PQ = 8m$$

Sol.45. (b)

$$\cos C = \frac{100 + 100 - 256}{2 \cdot 10 \cdot 10}$$

$$\cos C = \frac{-56}{200} = -\frac{7}{25}$$

$$\sin C = \frac{24}{25}$$

$$\frac{AB}{\sin C} = \frac{16}{24/25} = \frac{50}{3}$$

Sol.46. (d)

triangle is isosceles

angle A = angle B

$$\cos A + \cos B + \cos C$$

$$= 2 \cos A + \cos C$$

$$\cos A = \frac{100 + 256 - 100}{2 \cdot 10 \cdot 16} = \frac{4}{5}$$

$$2 \cos A + \cos C = \frac{8}{5} - \frac{7}{25} = \frac{33}{25}$$

What is $\cos A + \cos B + \cos C$ equal to?

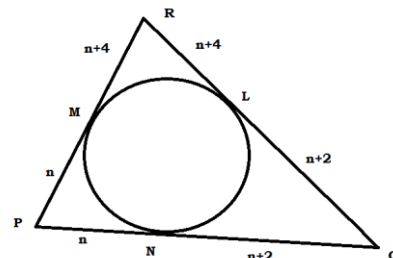
Sol.47. (a)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{16 + 9 - 4}{24} = \frac{7}{8}$$

$$\cos 3C = 4 \cos^3 C - 3 \cos C$$

$$= 4 \left(\frac{343}{512} \right) - 3 \left(\frac{7}{8} \right) = \frac{7}{128}$$

Sol.48. (c)



$$\cos P = \frac{PR^2 + PQ^2 + QR^2}{2 \cdot PR \cdot PQ}$$

$$\frac{1}{3} = \frac{(2n+4)^2 + (2n+2)^2 + (2n+6)^2}{2(2n+4)(2n+2)}$$

By solving it we will get $n = 8$

Sol.49. (d)

If $n=8$, then $PR = 20$, $PQ = 18$, $QR = 22$

Smallest side will be $PQ = 18$ unit

Sol.50. (c)

$$BC = a = \sqrt{3}$$

$$CA = b = 1$$

Perimeter = $6 \times \text{AM of sine of angles}$

$$a + b + c = 6 \times \frac{\sin A + \sin B + \sin C}{3}$$

$$a + b + c = 2 \times (\sin A + \sin B + \sin C)$$

$$[\text{we know that } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k]$$

$$\text{and } \sin A = ak, \sin B = bk, \sin C = ck]$$

$$a + b + c = 2 \times (ak + bk + ck)$$

$$a + b + c = 2 \times k(a + b + c)$$

$$1 = 2k$$

$$K = \frac{1}{2}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \frac{1}{2}$$

$$\frac{\sin A}{\sqrt{3}} = \frac{\sin B}{1} = \frac{\sin C}{c} = \frac{1}{2}$$

$A = 60^\circ$ and $B = 30^\circ$, then $C = 90^\circ$

$$\frac{\sin 90}{c} = \frac{1}{2}, \text{ so } c = 2$$

$$\text{Perimeter} = 2 + 1 + \sqrt{3} = 3 + \sqrt{3}$$

Sol.51. (c)

By above solution $C = 90^\circ$ so triangle is right angle

And angles of triangle are in AP

Sol.52. (c)

$$a^2 + b^2 + c^2 = ac + \sqrt{3}bc$$

$$a^2 + b^2 + c^2 - ac - \sqrt{3}bc = 0$$

$$a^2 - ac + \frac{c^2}{4} + b^2 - \sqrt{3}bc + \frac{3c^2}{4} = 0$$

$$\left(a - \frac{c}{2}\right)^2 + \left(b - \frac{\sqrt{3}}{2}c\right)^2 = 0$$

$$a - \frac{c}{2} = 0 \Rightarrow c = 2a \dots\dots\dots(1)$$

$$b - \frac{\sqrt{3}}{2}c = 0 \Rightarrow c = \frac{2b}{\sqrt{3}} \dots\dots\dots(2)$$

By (1) and (2)

$$2a = \frac{2b}{\sqrt{3}} = c \text{ or } \frac{a}{1/2} = \frac{b}{\sqrt{3}/2} = \frac{c}{1}$$

By comparing $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$$\sin A = \frac{1}{2} \text{ so } A = 30$$

$$\sin C = 1 \text{ so } C = 90$$

$$\text{and } B = 60$$

so triangle is right angle triangle

Sol.53. (c)

by using above solution

$$\frac{a}{1/2} = \frac{b}{\sqrt{3}/2} = \frac{c}{1}$$

$$\text{If } c = 8 \text{ then } a = 4 \text{ and } b = 4\sqrt{3}$$

$$\text{Area of right angle triangle} = \frac{1}{2} \times a \times b$$

$$= \frac{1}{2} \times 4 \times 4\sqrt{3} = 8\sqrt{3}$$

Sol.54. (c)

The angles A, B and C of a triangle ABC are in the ratio 3:5:4.

$$\text{Then } A = 45^\circ, B = 75^\circ, C = 60^\circ$$

What is the value of $a + b + \sqrt{2}c$

$$\frac{\sin 45}{a} = \frac{\sin 75}{b} = \frac{\sin 60}{c} = \frac{1}{k}$$

$$\frac{1}{\sqrt{2}a} = \frac{\sqrt{3}+1}{2\sqrt{2}b} = \frac{\sqrt{3}}{2c} = \frac{1}{k}$$

$$a = \frac{k}{\sqrt{2}}, b = \frac{\sqrt{3}+1}{2\sqrt{2}}k, c = \frac{\sqrt{3}}{2}k$$

$$a + \sqrt{2}c = \frac{k}{\sqrt{2}} + \frac{\sqrt{3}}{2}k = \left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)k = 2b$$

$$a + b + \sqrt{2}c = 3b$$

Sol.55. (a)

$$a = \frac{k}{\sqrt{2}}, b = \frac{\sqrt{3}+1}{2\sqrt{2}}k, c = \frac{\sqrt{3}}{2}k$$

$$a^2 = \frac{k^2}{2}, b^2 = \frac{(\sqrt{3}+1)^2}{8}k^2, c^2 = \frac{3}{4}k^2$$

$$a^2 : b^2 : c^2$$

$$= \frac{k^2}{2} : \frac{(\sqrt{3}+1)^2}{8}k^2 : \frac{3}{4}k^2$$

$$= \frac{1}{2} : \frac{4+2\sqrt{3}}{8} : \frac{3}{4}$$

$$= 2 : 2 + \sqrt{3} : 3$$

Sol.56. (a)

In a triangle ABC, $AB = 16$ cm, $BC = 63$ cm and $AC = 65$ cm. What is the value of $\cos 2A + \cos 2B + \cos 2C$?

$$\text{we know that } 16^2 + 63^2 = 65^2$$

so triangle is right angled at angle B

$$B = 90^\circ \text{ and } A + C = 90^\circ$$

$$\cos 2A + \cos 2C + \cos 2B$$

$$\Rightarrow 2 \cos(A+C) \cos(A-C) + \cos 2B$$

$$\Rightarrow 2 \cos 90^\circ \cos(A-C) + \cos 180^\circ$$

$$\Rightarrow 0 - 1 = -1$$

Sol.57. (a)

$$\cot A \cdot \cot B \cdot \cot C > 0$$

i.e. all angles are acute

if any one angle become obtuse than cotangent of that angle will be negative.

so all angles are acute.

$$\cot A \cdot \tan B \cdot \tan C > 0$$

i.e. all angles are acute

statement 1 is correct

if any one angle become obtuse than tangent of that angle will be negative.

so all angles are acute.

statement 2 is incorrect

Sol.58.

$$\frac{\cos A + \cos B}{\cos\left(\frac{A-B}{2}\right)} = \frac{2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A-B}{2}\right)}$$

$$C = 60^\circ \text{ so } A + B = 120^\circ$$

$$2 \cos\left(\frac{120}{2}\right) = 1$$

Sol.59. (a)

$$\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C} \dots\dots(i)$$

but we know that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \dots\dots(ii)$$

from (i) and (ii)

$$\frac{a}{b} = \frac{\sin A}{\sin B} = \frac{\cos A}{\cos B}$$

$$\text{i.e. } \tan A = \tan B$$

same as above $\tan B = \tan C$

if $\tan A = \tan B = \tan C$

$$\text{then } A = B = C$$

triangle is equilateral

Area of equilateral triangle

$$= \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} (6)^2 = 9\sqrt{3}$$

Sol.60. (c)

roots of the equation $7x^2 - 6x + 1 = 0$ are $\tan \alpha$ and $\tan \beta$

$$\tan \alpha + \tan \beta = \frac{6}{7} \text{ and } \tan \alpha \cdot \tan \beta = \frac{1}{7}$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\frac{6}{7}}{1 - \frac{1}{7}} = 1$$

$$\text{so } \alpha + \beta = 45^\circ$$

$$\text{and } 2\alpha + 2\beta = 90^\circ$$

in question it is given that 2α and 2β are the angles of triangle so triangle is right angled triangle.

from any statements or result we cannot conclude that $\alpha = \beta$, so triangle is not isosceles.

Sol.61. (b)

we know that

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 75}{a} = \frac{\sin 45}{b} = \frac{\sin 60}{c}$$

$$\frac{a}{\sin 75} = \frac{b}{\sin 45} = \frac{c}{\sin 60}$$

$$\frac{\sqrt{3}+1}{2\sqrt{2}a} = \frac{1}{\sqrt{2}b} = \frac{\sqrt{3}}{2c}$$

$$\frac{\sqrt{3}+1}{2\sqrt{2}a} = \frac{1}{\sqrt{2}b} = \frac{\sqrt{3}}{2c}$$

$$\frac{\sqrt{3}+1}{2\sqrt{2}a} = \frac{1}{\sqrt{2}b} = \frac{\sqrt{3}}{2c}$$

by above relation

$$2a = \frac{(\sqrt{3}+1)}{\sqrt{3}}\sqrt{2}c \text{ and } b = \frac{\sqrt{2}}{\sqrt{3}}c$$

$$2a - b = \frac{(\sqrt{3}+1)}{\sqrt{3}}\sqrt{2}c - \frac{\sqrt{2}}{\sqrt{3}}c = \sqrt{2}c$$

Sol.62. (b)

for a triangle we know that

$$A + B + C = 180^\circ$$

$$A + B = 180^\circ - C$$

$$\tan(A + B) = -\tan C$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$1 - \tan A \tan B$$

$$\tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\text{given that } \tan A + \tan B + \tan C = k$$

$$\text{so } \tan A \tan B \tan C = k$$

$$\text{and } \cot A \cdot \cot B \cdot \cot C = 1/k$$

Sol.63. (c)

side CA is longest so its opposite angle B will be largest

$$\cos B = \frac{AB^2 + BC^2 - CA^2}{2AB \cdot BC}$$

$$= \frac{9 + 25 - 49}{2 \cdot 3 \cdot 5} = -\frac{1}{2}$$

$$\cos B = -\frac{1}{2} \Rightarrow B = 120^\circ$$

So triangle is obtuse angled triangle and sum of other two angles is 60° i.e. acute.

Sol.64. (c)

By above solution

$$B = 120^\circ$$

Sol.65. (a)

$$\text{Area of triangle} = \frac{1}{2} \times AB \times BC \times \sin B$$

$$= \frac{1}{2} \times 3 \times 5 \times \sin 120^\circ = 15\sqrt{3}/4 \text{ sq. cm}$$

Sol.66. (b)

The angles A, B of a triangle ABC are in the ratio 3 : 1

Let angle A is $3x$ and B is x

$$\frac{\sin A}{a} = \frac{\sin B}{b} \Rightarrow \frac{\sin 3x}{2k} = \frac{\sin x}{k}$$

$$\sin 3x = 2 \sin x$$

$$3 \sin x - 4 \sin^3 x = 2 \sin x$$

$\sin x = 1/2$
 $x = 30^\circ$
angle $x = B = 30^\circ$,

and $A = 3x = 90^\circ$, $C = 60^\circ$
Sol.67. (c)
by above solution
triangle is right angled triangle

angle $B = 30^\circ$, $A = 90^\circ$, $C = 60^\circ$
so angles are in AP

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